

Vivek Jagadeesh

508-243-5976 | vivekjag1@gmail.com | vivekjagadeesh.com | linkedin.com/in/vivek-jagadeesh | github.com/vivekjag1

EDUCATION

Worcester Polytechnic Institute (WPI)

Worcester, MA

Master of Science in Computer Science (Thesis Track), GPA: 4.0/4.0

Graduation May 2027

Bachelor of Science in Computer Science – Concentration in Cybersecurity, GPA: 3.97/4.0

Graduated May 2026

SKILLS

Cybersecurity & Systems: System Hardening, Vulnerability Research, Kernel Development, OS, Cryptography

Programming Languages: Rust, C, C++, Python, Java, SQL, JavaScript, TypeScript

Emulation & Development Tools: Linux, Docker, QEMU, Git, VirtualBox

Databases & Cloud: PostgreSQL, Oracle, SQLite, AWS (EC2, RDS, S3, DynamoDB, Lambda), Azure (Cosmos DB)

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, Flask, GraphQL, Tailwind CSS, Prisma ORM, Auth0

Machine Learning: Scikit-learn, AWS Bedrock, Retrieval Augmented Generation (RAG), Agent Flows, OpenSearch

EXPERIENCE

Software Engineering Intern

Jun 2025 – Aug 2025

Waters Corporation

Milford, MA

- Engineered and deployed a Retrieval Augmented Generation (RAG) chatbot to streamline order support
- Developed a secure user authentication system using AWS Cognito and token verification system
- Created serverless architecture using 7 AWS Lambda Functions for response generation and document retrieval
- Designed WebSocket layer to enable real-time communication between UI and Lambda Functions
- Implemented an AI Agent Flow to allow customers to re-order previous purchases directly from the chatbot
- Presented technical demonstrations to the senior leadership team and global stakeholders

PROJECTS

Secure Memory Allocator for the Linux Kernel, Major Qualifying Project

Aug 2025 – Present

- Developing a secure memory allocator for the Linux Kernel using ARM Memory Tagging Extension (MTE)
- Conducted in-depth vulnerability research into modern Linux Kernel exploits, specifically DirtyCred
- Architected a custom kernel module and 3 new system calls to create interface between user space and allocator
- Cross-compiled, tested, and deployed the modified Linux kernel onto an embedded hardware target (Pixel 9A)
- Established testing framework using KUnit and custom benchmarks to measure security and performance
- Testing allocator stability and security using QEMU, Docker, and AOSP
- Awarded WPI's Provost's MQP Award for outstanding undergraduate research

MLB Pitch Classification System, Personal Project

July 2024 – Aug 2024

- Developed machine learning system to classify MLB pitches from statistical data using Scikit-learn and Python
- Selected features from the dataset to tune and train a neural network, achieving a predictive accuracy of 92%
- Connected model with a custom web application using Next.js, Python, Flask, SciKit-Learn and PostgreSQL
- Implemented a CI/CD pipeline using GitHub actions and Docker to automate the containerization process

An application for Brigham and Women's Hospital, CS 3733: Software Engineering

Mar 2024 – May 2024

- Contributed 8,500 lines of code and 128 commits as Assistant Lead Software Engineer for a 10-person team
- Engineered a security system with Auth0, featuring RBAC, automated user provisioning, and protected APIs
- Developed and maintained the system's backend, including Express.js REST APIs and PostgreSQL database
- Deployed application to AWS EC2 and RDS through CI/CD pipeline using Docker and GitHub actions
- Led scrum meetings, retrospectives, technology stack presentations, and two application demonstrations
- Created and implemented entity-relationship and UML use case diagrams to simplify architectural decisions

LEADERSHIP & TEACHING EXPERIENCE

Student Teaching Assistant

May 2024 – May 2026

Worcester Polytechnic Institute

Worcester, MA

- Mentored 24 students across 6 agile teams in design, development, and cloud deployment of web applications
- Led recitation sessions on cloud deployment and containerization using AWS EC2, RDS, and Docker
- Conducted weekly code reviews and provided actionable feedback to enforce software engineering standards
- Guided teams in implementing agile ceremonies (Scrum, retrospectives, backlog refinement)